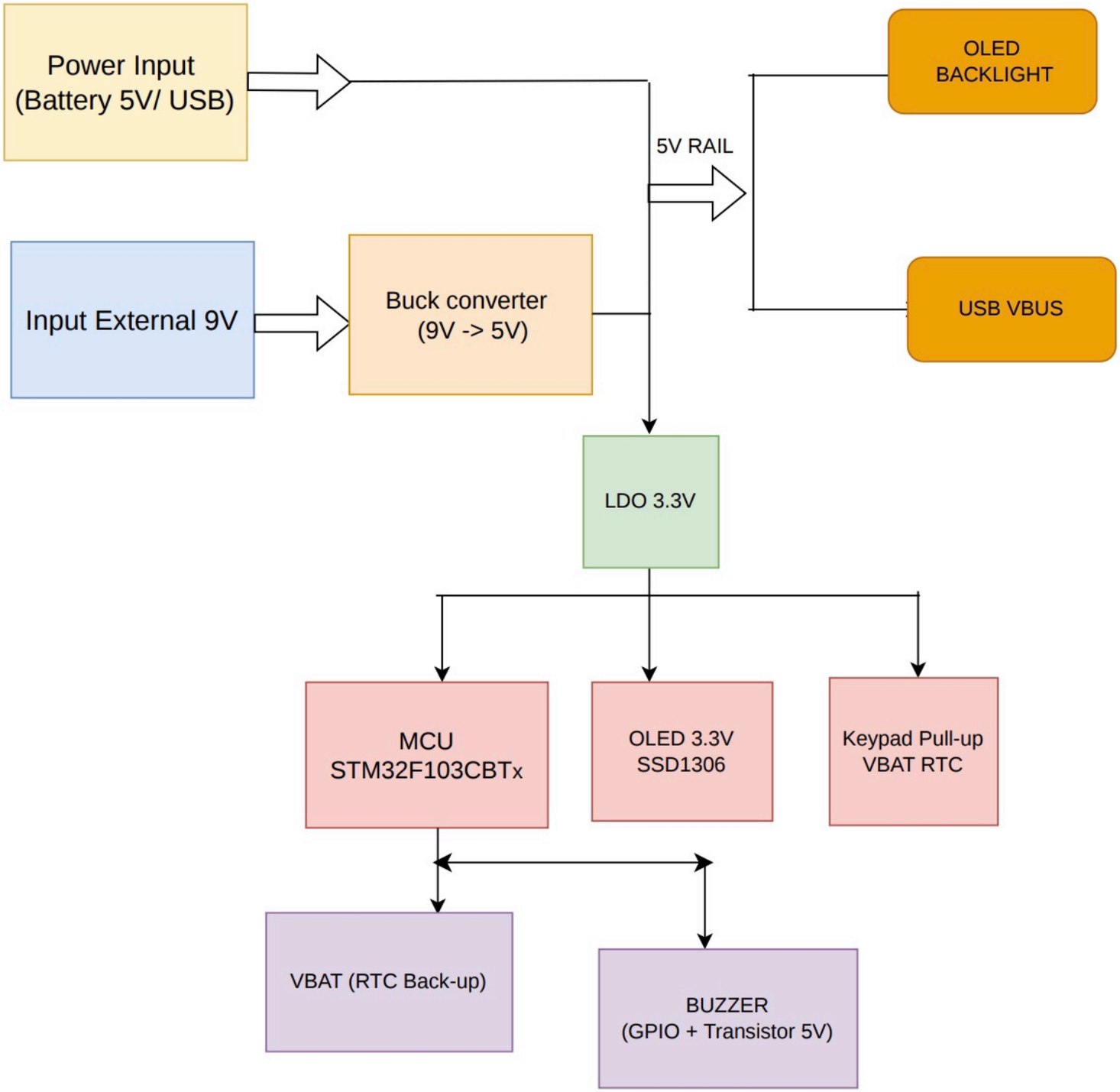

Power Input
(Battery 5V/ USB)
OLED
BACKLIGHT
5V RAIL
Input External 9V
Buck converter
(9V -> 5V)
USB VBUS
LDO 3.3V
MCU
STM32F103CBTx
OLED 3.3V
SSD1306
Keypad Pull-up
VBAT RTC
VBAT (RTC Back-up)
BUZZER
(GPIO + Transistor 5V)

BLOCK DIAGRAM – CALCULATOR

POWER INPUT
(USB 5V / Battery)

PROTECTION & FILTERING
D1 (reverse)
D_Zener (OVP)
C1, C3, C4 (filter)

LDO 3.3V
(AMS1117-3.3)
IN —— C5 (10µF) ⏚ GND
OUT —— C6 (10µF), C7 (0.1µF) ⏚ GND

3.3V (VDD_MAIN)

MCU BLOCK (STM32)

POWER DECOUPLING
C8..C17 (0.1µF each)
near each VDD/VDDA/Vref+

BOOT SELECTOR
SW1 (push) --- R1 (10k) GND
R_US (10k) --- VDD_MAIN
BOOT0 pin

VBAT / RTC
D1_Schottky (from VCC)
D2_Schottky (from CR2032)
D_Zener (3.6V)
C_VBAT (1uF)
VBAT pin (MCU)

KEYPAD
(4x4 matrix)
Rows: PA0-PA3
Cols : PB0-PB3
Pull-ups (10k) optional

LCD (I2C)
(16x2 + PCF8574)
SDA ----- PB6
SCL ----- PB7
VDD --- VDD_MAIN (3.3V)
GND --- GND
Pull-ups (4.7k) on I2C

BUZZER (opt)
GPIO (PC13) --- R(1k) --- Base NPN
Collector --- Buzzer --- VDD_MAIN
Emitter --- GND

FIRMWARE ARCHITECTURE

APPLICATION LAYER

Calculator Logic

- State machine (idle, input, result, error)
- Expression evaluation (postfix / stack)
- Display formatting

SERVICE LAYER

Keypad Service

- Scan
- Debounce
- Event

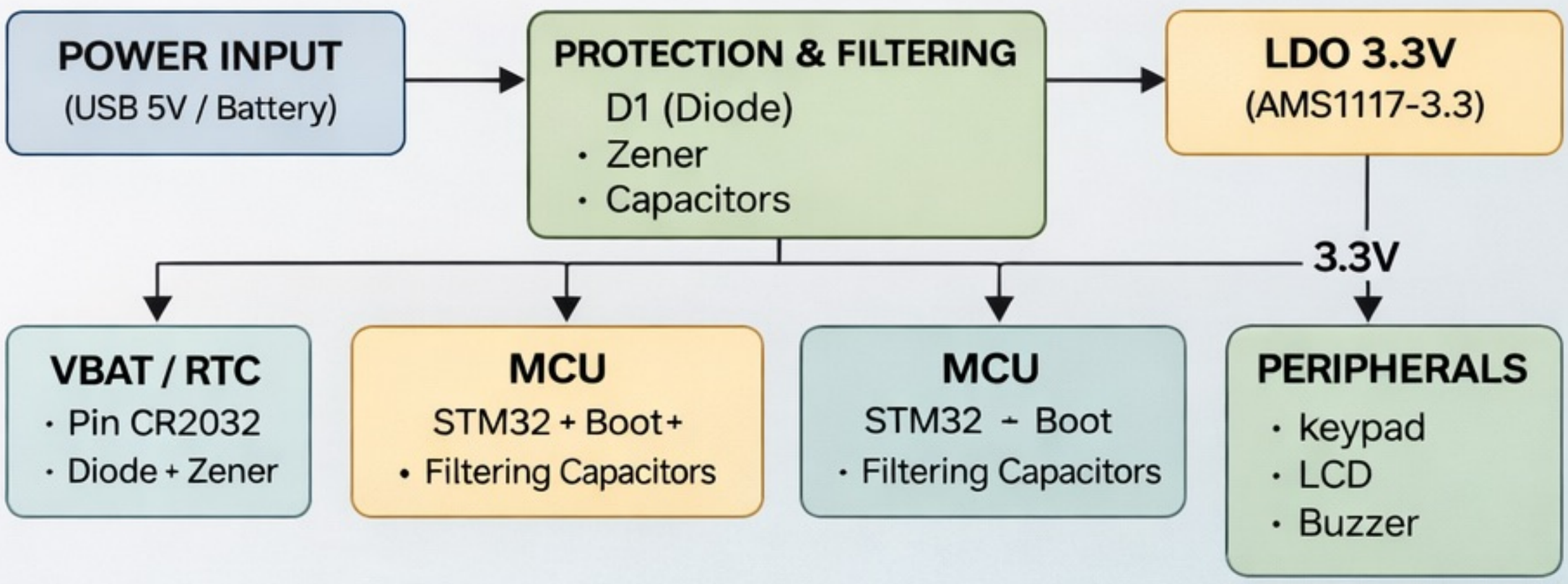
LCD Service

- Write
- Clear
- Cursor

Buzzer Service

- Beep
- (short, long)

HAL (HARDWARE ABSTRACTION LAYER)

- GPIO (input, output, interrupt)
- I2C (LCD)
- Timer (keypad scanning, debouncing, buzzer timing)
- RTC (if used)

SYSTEM POWER OVERVIEW